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Key Points:

- Peak X-ray and ENA emission occurs along tangent line of sight
- Series of tangent observations allows surface reconstruction
- Algorithm reconstructs magnetopause from tangent observations

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Magnetopause Surface Reconstruction From Tangent Vector Observations

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Abstract Entire fields of science, most notably in astrophysics, rely on line-of-sight observations. In planetary science and heliophysics, the techniques of soft X-ray and energetic neutral atom imaging also produce line-of-sight measurements. An important question is whether the geometry of the surface, for example, the magnetopause, can be reconstructed using only line-of-sight observations from a single spacecraft. Under a broad range of conditions, the peak emission corresponds to the tangent to the boundary surface, such as the planetary surface or magnetopause, the so-called *limb brightening* phenomenon. Thus, line-of-sight observations frequently provide information concerning the tangent to the surfaces being observed. We present an algorithm to reconstruct the cross section of the magnetopause using line-of-sight soft X-ray observations (and, in principle, energetic neutral atom observations). The algorithm successfully reconstructs the cross section of the magnetopause in the orbit plane. The three-dimensional magnetopause structure can be recovered from observations by a spacecraft whose orbit precesses around the magnetosphere.

1. Introduction

The tangent vector plays a central role in differential geometry; for example, it appears in the expressions for the Frenet apparatus (e.g., O'Neill, 1966), which defines a curve only with respect to properties of the curve itself. On a more practical level, tangent vectors can reveal global geometry. For example, ancient seafaring people such as the Greeks realized the Earth must be curved because they could see tall masts of receding ships long after their hulls had disappeared.

In the modern era, remote observations in astrophysics, heliophysics, and planetary science rely on line-of-sight observations which frequently show the greatest emission strength when the line-of-sight runs tangent to an observed surface, the so-called limb brightening phenomenon (e.g., Collier et al., 2005, 2014). In astrophysics, limb brightening occurs near a celestial body surrounded by an optically thin photon-emitting region. The look angle that touches the edge of the body has the longest path length through the emission region. More emission occurs along this direction, producing a bright edge. This limb brightening effect is particularly prominent at ultraviolet wavelengths and has been used, for example, to study properties of the exoplanet HD209458b (Schlawin et al., 2010). Limb brightening in this stellar context is an empirically observed phenomenon that in the optically thin case occurs because the chromospheric and transition region gas has its largest column density at the edges. In the case of limb brightening due to solar wind charge exchange at the Moon, Collier et al. (2014, section 8) show that limb brightening increases the observed column density by a factor of 10 above that along the radial line of sight, so that the observation geometry dominates the emission profile. Thus, the tangent vector to the observed surface may in general be the easiest physical parameter to deduce from remote observations. In fact, the upcoming European Space Agency-Chinese Academy of Sciences Solar Wind-Magnetosphere-Ionosphere Link Explorer (SMILE) will perform soft X-ray imaging of the magnetosheath.

This paper presents an algorithm for reconstructing the surface of an object from a set of tangent vector direction observations assuming time stationarity. There are periods over which solar wind conditions remain steady over long times. For example, Fuselier et al. (2010) averaged over an 11-hr period to generate an energetic neutral atom (ENA) image of the subsolar magnetosheath. During this period, the solar wind speed varied by only about $\pm 1.8\%$ and the solar wind density varied by only about $\pm 12\%$, resulting in a subsolar magnetopause position that was stable to within 0.2 Earth radii.

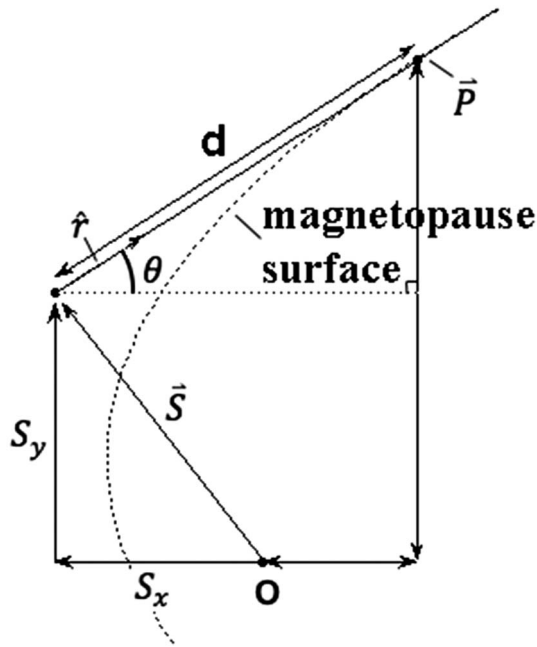


Figure 1. Variables for the derivation of the general formula for the reconstruction of the surface from spacecraft position, $\vec{s}$, parametrized by tangent angle, θ .

Alternatively, as we discuss later, satellite data could be binned for similar conditions producing an effective time stationarity for the data set. We also assume that the position, $\vec{s}$, of the observing spacecraft can be parametrized by the tangent vector angle, θ , and that we know how the spacecraft's position varies with the line-of-sight direction, that is, that we know $d\vec{s}/d\theta$. That this can be done is not surprising because the tangent vector is the derivative of the curve. Thus, reproducing a curve from tangent vector observations is analogous to reproducing a function based on knowledge of its derivatives.

Since at least the early 1990s within the field of computer vision, researchers have developed numerous so-called Structure from Motion (SfM) techniques. These are modeling methods using representative image points in a sequence of two-dimensional images to estimate three-dimensional structures (e.g., Tomasi & Kanade, 1992).

These SfM approaches are fundamentally different from the algorithm we describe here. SfM algorithms are photogrammetric range imaging techniques, meaning they are designed to employ photographs to recover the location of surface points. As such, the standard implicit assumptions in these algorithms are that the object imaged is a solid rigid object and that the images contain many intrinsic feature points that can be tracked from one image to the next (see, e.g., Skarbek, 2014; Figures 1–3).

Soft X-ray and ENA imaging, by contrast, are line-of-sight imaging techniques that sample not an object itself (e.g., the magnetopause) but the distribution of plasma and thermal neutral atom properties surrounding that object. Thus, the object's apparent morphology can depend critically on the observer's perspective varying with the plasma and neutral atom characteristics along the observer's line of sight (unlike, e.g., imaging a face that looks like a face regardless of the observer's location). Furthermore, the types of features typically tracked from image to image in SfM such as corner points that have different gradients in different directions only exist at thin boundaries like shocks in space plasmas, and these structures are not directly observed by soft X-ray or ENA imaging.

2. Derivation of the Reconstruction Equation

Figure 1 illustrates the geometry for the derivation of the reconstruction equation. Here the dotted line is the curve we are reconstructing based on a series of tangent vector observations taken during the motion of a spacecraft. This curve is the cross section of a three-dimensional surface in the plane of the spacecraft motion, which we will think of as the magnetopause but in principle can be any surface from which we can acquire tangent observations. The point **O** is an arbitrary origin that we are free to establish wherever we like. $\hat{r}$ is the line-of-sight tangent direction unit vector, which is the unit vector along the direction of the tangent line of sight. $\vec{s}$ is the vector defining the spacecraft location relative to the arbitrarily located origin, **O**. θ is then the observed angle between the line-of-sight tangent and the positive x direction.

Now, consider a coordinate system, as shown in Figure 2, in which the x axis is along the line-of-sight direction, $\hat{r}$. The magnetopause surface is the thick solid line, and the original tangent point is labeled P whose coordinates are (x_0, y_0) . The spacecraft then moves a small amount, $\Delta\vec{s}$, to a new position where the tangent angle with the x axis is now $\Delta\theta$ and the new tangent point position has coordinates (x_1, y_1) . Because we are considering small changes, we can describe the magnetopause surface $y(x)$ around point (x_0, y_0) by a Taylor series in which we retain terms of up to second order:

$$y(x) = y_0 + \frac{1}{2} \frac{d^2y}{dx^2} \bigg|_{x_0} \cdot (x - x_0)^2, \quad (1)$$

where the first-order term in the series disappears because the point P is an extremum, and the first derivative there is 0.

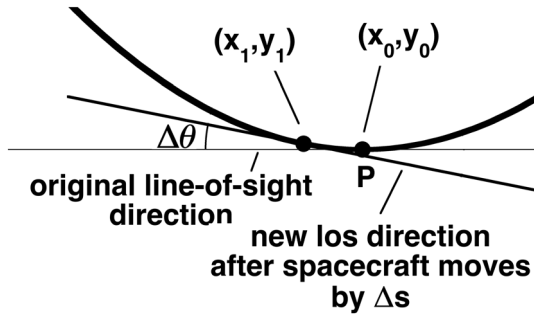


Figure 2. The tangent point position varies only along the line-of-sight direction $\hat{r}$ for small changes in spacecraft position.

When the spacecraft moves to its new location a small distance away and observes the tangent point to now be at (x_1, y_1) , the slope of this new line-of-sight direction can be determined by taking the derivative of equation (1) and evaluating it at $x = x_1$:

$$\left. \frac{dy}{dx} \right|_{x_1} = \left. \frac{d^2y}{dx^2} \right|_{x_0} \cdot (x_1 - x_0) = \left. \frac{d^2y}{dx^2} \right|_{x_0} \Delta x, \quad (2)$$

where we have denoted the change in the position of the tangent point by Δx . Now, the slope of the new tangent line is also the tangent of its angle with respect to the x axis so

$$\tan(\Delta\theta) = \left. \frac{dy}{dx} \right|_{x_1}, \quad (3)$$

where, unlike at x_0 , the first derivative is nonzero at x_1 . Because the angle $\Delta\theta$ is small, we can write $\tan(\Delta\theta) = \Delta\theta$, so the relationship between the change in position of the tangent point, Δx , and the change in the angle of the tangent direction, $\Delta\theta$, is given by

$$\Delta\theta = \left. \frac{d^2y}{dx^2} \right|_{x_0} \Delta x. \quad (4)$$

Equation (4) can now be combined with equation (1) evaluated at the point x_1 to show that the position of the new tangent point is given by

$$y(x_1) = y_0 + \frac{(\Delta\theta)^2}{2 \left. \frac{d^2y}{dx^2} \right|_{x_0}}, \quad (5)$$

so that a first-order change in the tangent angle, $\Delta\theta$, results in a second-order change in the tangent point position ($y(x_1) - y_0$).

In the coordinate system shown in Figure 2, the direction along the line of sight, the x axis, is $\hat{r}$ and the direction perpendicular to the x axis, the y axis, is $\hat{\theta}$. Thus, the point (x_0, y_0) is defined by the vector $\vec{p}_0 = x_0\hat{r} + y_0\hat{\theta}$ and the point (x_1, y_1) is defined by the vector $\vec{p}_1 = x_1\hat{r} + y_1\hat{\theta}$, and

$$\Delta\vec{p} = \vec{p}_1 - \vec{p}_0 = (x_1 - x_0)\hat{r} + (y_1 - y_0)\hat{\theta} = \Delta x\hat{r} + \Delta y\hat{\theta}. \quad (6)$$

Using equation (5), the change in position of point $\vec{p}$ in the coordinate system in Figure 2 becomes

$$\Delta\vec{p} = \Delta x\hat{r} + \frac{(\Delta\theta)^2}{2 \left. \frac{d^2y}{dx^2} \right|_{x_0}} \hat{\theta}. \quad (7)$$

Dividing both sides of equation (7) by $\Delta\theta$, we get

$$\frac{\Delta\vec{p}}{\Delta\theta} = \frac{\Delta x}{\Delta\theta} \hat{r} + \frac{\Delta\theta}{2 \left. \frac{d^2y}{dx^2} \right|_{x_0}} \hat{\theta}. \quad (8)$$

Taking the limit of equation (8) as $\Delta\theta$ goes to 0, we arrive at

$$\frac{d\vec{p}}{d\theta} = \frac{dx}{d\theta} \hat{r}, \quad (9)$$

where the $\hat{\theta}$ term goes to 0. Equation (9) shows that the vector $\vec{p}$ only varies along the line of sight, that is, along $\hat{r}$, and does not vary perpendicular to the line of sight, in the $\hat{\theta}$ direction, because the point P is a tangent point.

Referring back now to Figure 1, the vector $\vec{p}$ representing the tangent point on the magnetopause surface is the sum of the spacecraft position vector, $\vec{s}$, and a vector of length d along the line-of-sight direction, $\hat{r}$, or formally

$$\vec{p} = d\hat{r} + \vec{s}. \quad (10)$$

Taking the derivative of equation (10) with respect to θ and, using equation (9) to set it equal to $dx/d\theta\hat{r}$, we get

$$\frac{d\vec{p}}{d\theta} = \frac{d\vec{s}}{d\theta} + \frac{dd}{d\theta}\hat{r} + d\frac{d\hat{r}}{d\theta} = \frac{dx}{d\theta}\hat{r}. \quad (11)$$

Now we note with reference to Figure 1 that

$$\hat{r} = \cos\theta\hat{x} + \sin\theta\hat{y}. \quad (12)$$

Then if $\hat{\theta}$ is the vector perpendicular to $\hat{r}$ in the direction in which the angle θ increases, then

$$\hat{\theta} = -\sin\theta\hat{x} + \cos\theta\hat{y}. \quad (13)$$

From equations (12) and (13), we conclude that

$$\frac{d\hat{r}}{d\theta} = -\sin\theta\hat{x} + \cos\theta\hat{y} = \hat{\theta}, \quad (14)$$

so that equation (11) becomes

$$\frac{d\vec{p}}{d\theta} = \frac{d\vec{s}}{d\theta} + \frac{dd}{d\theta}\hat{r} + d\hat{\theta} = \frac{dx}{d\theta}\hat{r}. \quad (15)$$

Now we take the dot product of both sides of equation (15) with $\hat{\theta}$ and note that $\hat{\theta}$ and $\hat{r}$ are basis vectors so, by definition, $\hat{r}\cdot\hat{\theta} = 0$ and $\hat{\theta}\cdot\hat{\theta} = 1$. Thus,

$$0 = \frac{d\vec{s}}{d\theta} \cdot \hat{\theta} + d, \quad (16)$$

or

$$d = -\frac{d\vec{s}}{d\theta} \cdot \hat{\theta} = \left| \frac{d\vec{s}}{d\theta} \cdot \hat{\theta} \right|, \quad (17)$$

because distance d is positive definite.

Substituting this expression back into the equation (10), $\vec{p} = \vec{s} + d\hat{r}$, we get

$$\vec{p} = \left| \frac{d\vec{s}}{d\theta} \cdot \hat{\theta} \right| \hat{r} + \vec{s}, \quad (18)$$

which is the general formula for reconstruction of the surface from the spacecraft position, $\vec{s}$, parametrized by tangent angle, θ .

For a qualitative motivation of why this algorithm successfully reconstructs the magnetopause surface, consider Figure 3. A spacecraft at point SC1 only observes the direction in its field of view of the magnetopause surface. So, for example, a spacecraft at point SC1 cannot determine whether the magnetopause is at point MP1 on surface 1 or at point MP2 on surface 2. However, by determining how the observation angle changes as the spacecraft moves to another very close point SC2, the ambiguity can be removed, as shown in Figure 3.

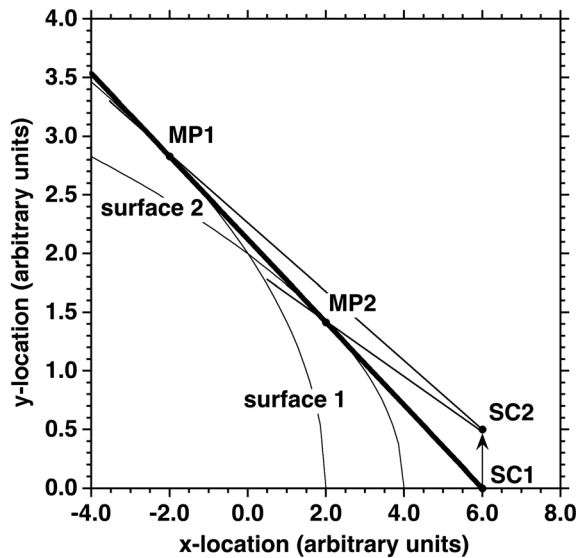


Figure 3. From a single location, there exists ambiguity in the location of the magnetopause. The motion of the spacecraft removes this ambiguity.

So the magnetopause location can be determined by the derivative of the spacecraft position with respect to the angle of the tangent direction as expressed quantitatively by equation (18).

3. Test of Reconstruction Algorithm Using Soft X-Ray Simulations—Orbits in the GSE YZ and XZ Planes

As shown in the preceding section, the magnetopause surface can be reconstructed based only on knowledge of the spacecraft position, the line-of-sight direction, and how the spacecraft position varies with the line-of-sight direction, that is, with the spacecraft position parametrized by that angle.

To determine the validity of the assumption that local peaks in the observed solar wind charge exchange soft X-ray flux coincide with the line-of-sight tangents to the surfaces, we have performed magnetohydrodynamic (MHD) simulations using OpenGGCM (e.g., Honkonen et al., 2013; Raeder et al., 2008). The MHD models were run and then line-of-sight integrations were performed to determine the total soft X-ray flux observed in that direction due to solar wind charge exchange with exospheric neutral hydrogen (e.g., Cravens et al., 2001).

The line-of-sight integrations were performed by first calculating the product of the exospheric neutral density, the plasma density, and the effective plasma speed, a quantity called Q_{cut} , at each point in the simulation grid. We then integrated Q_{cut} along each line of sight. The plasma density and effective plasma speeds are provided by the MHD simulation, and, to calculate the neutral density, n_H , we used the relationship

$$n_H = 25 \text{ cm}^{-3} \left(\frac{10}{R} \right)^3, \quad (19)$$

where R is the radial distance from the Earth's center measured in Earth radii, $R_E = 6371 \text{ km}$ (Cravens et al., 2001). This integrated value, called Q , is proportional to the emitted soft X-ray intensity (Kuntz et al., 2015). To calculate the soft X-ray flux from the MHD simulation, we assume that the density of high charge state ions, those that generate the soft X-rays when they interact with exospheric neutral atoms, is proportional to that of the protons, the ions tracked by single-species global MHD models.

Figure 4 shows an example of the results from such an OpenGGCM MHD simulation for a southward interplanetary magnetic field (IMF) of 5 nT, a solar wind density of 10 cm^{-3} , and a flow speed of 400 km/s. In this case, the spacecraft was orbiting in the geocentric solar ecliptic (GSE) YZ plane (GSE $X = 0$) at a distance of 40 Earth radii (R_E) and viewing the magnetopause surface in this plane.

These types of orbits, well outside of the magnetopause, are typical of proposed soft X-ray imaging missions, and the algorithm described is applicable to any orbit as long as it remains outside the magnetopause and therefore can observe the tangent to the magnetopause. In particular, SMILE will have a highly elliptical polar orbit with an apogee beyond the magnetopause, so this algorithm will be applicable to SMILE data.

The top left panel in Figure 4 shows the local soft X-ray emission at each point in this plane, Q_{cut} (this quantity is integrated along the line of sight to calculate the total emission, Q), and the top right panel shows the plasma density. Although densities peak in the outermost magnetosheath (just inside the bow shock), emissions peak in the inner magnetosheath (just outside the magnetopause). This occurs because the total observed intensity depends not only on plasma density but also on the exospheric neutral density and effective velocity.

The lines in these top two panels originate at the spacecraft location and show both the direction of the local peak in the soft X-ray emission (black) and the direction of the tangent to the magnetopause location (red). The magnetopause position along each radial line starting at the Earth's center is taken to be the location where the gradient in the plasma density reaches a peak (magenta; Garcia & Hughes, 2007). Note that

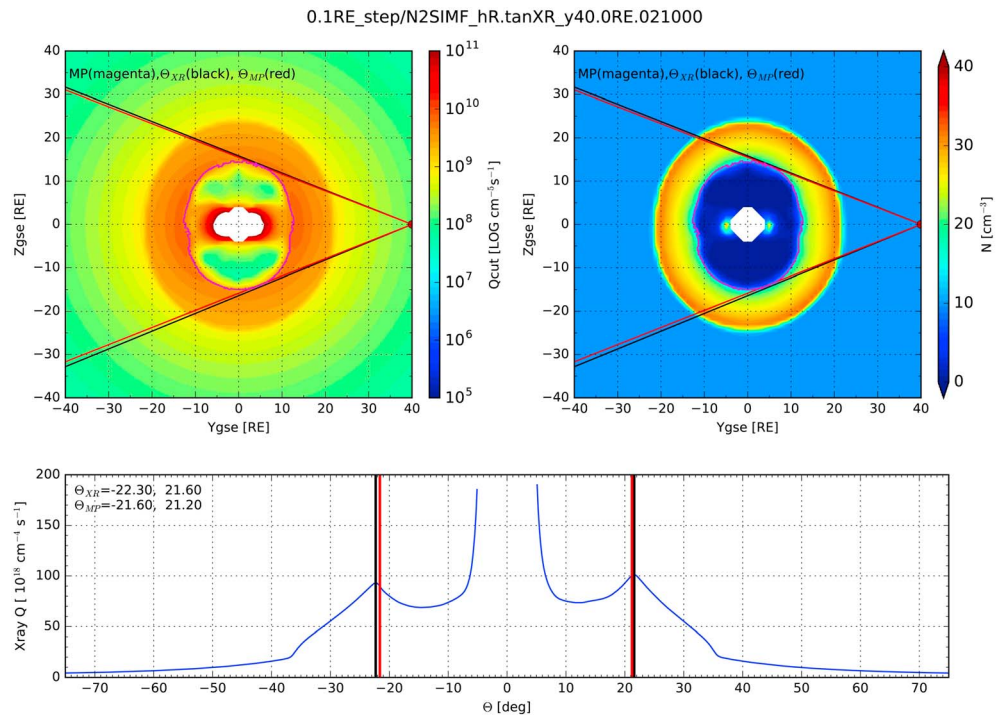


Figure 4. Correspondence between the peak soft X-ray emission and the tangent to the magnetopause. Note that the emission inside the magnetopause would not exist in an actual soft X-ray image because the charge state of that plasma is too low to emit in soft X-rays. The magnetopause position is taken to correspond to the maximum plasma density gradient.

because the magnetopause has some thickness, identifying the magnetopause position with the peak in the gradient results in a location with plasma density less than half the magnetosheath density. So the angle at which the soft X-ray emission peaks lies about $0.2\text{--}0.3 R_E$ sunward of the magnetopause location identified by the maximum in the plasma density gradient. Here the plasma density is typically over 80% of the magnetosheath density.

There are many methods for calculating the magnetopause location from MHD simulation output including the peak density gradient, the last closed field line, the maximum current density, and the plasma streamline techniques. Among these, the peak density gradient method is the most useful because it determines the magnetopause location reliably throughout a large region for all IMF conditions. The last closed field line technique cannot locate the high-latitude magnetopause because open field lines occupy this region. The maximum current density technique can be unreliable especially for northward IMF because the small magnetic shear near the low-latitude magnetopause produces a weak current density. The plasma streamline technique is inaccurate near the subsolar point because most plasma streamlines flow away from this point and never touch the subsolar magnetopause. In contrast, the peak density gradient method accurately locates the magnetopause at both low and high latitudes even for northward IMF. Because we simulate both southward and northward IMF conditions, the peak density gradient method is the most appropriate for this study.

The lower panel in Figure 4 shows that the direction of the peak emission, θ_{XR} , and the direction of the tangent to the magnetopause, θ_{MP} , coincide to within less than a degree by plotting the line-of-sight integrated emission versus the line-of-sight angle. As expected, the difference between the two angles increases as the observation point gets closer to Earth. For the data shown in Figure 4, the angle increases from 0.4° to 3.5° as the spacecraft moves from 40 to $15 R_E$. Note that there are clear local peaks in the line-of-sight emission corresponding to the direction of the magnetopause tangent. This appears to be a general property, namely, that a local peak in the observed soft X-ray flux occurs coincident with the geometric tangent line of sight to the surface.

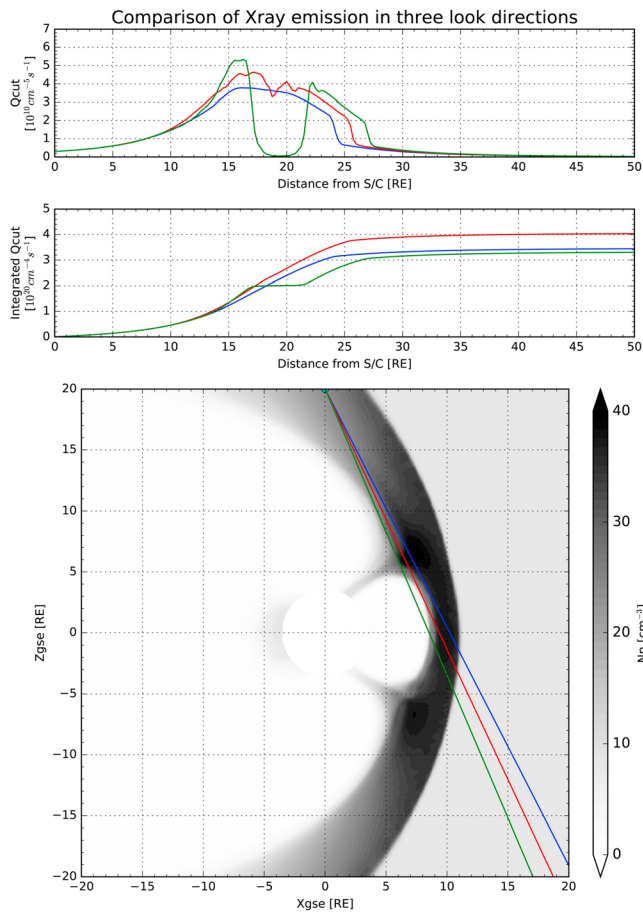


Figure 5. A comparison of soft X-ray emission along three line of sights.

This is not entirely surprising because, as described above, the soft X-ray emission tends to be strongest at the point closest to the center of the Earth, namely, at the magnetopause. Although the peak soft X-ray emission depends on the product of the exospheric density, the plasma density, and the effective velocity of the highly charged solar wind ions, the exospheric density increases very rapidly as the distance from Earth's center decreases (e.g., Collier et al., 2005). This effect tends to dominate the emission so the greatest emission frequently occurs where the exospheric density is highest, that is near the magnetopause. Also, the tangent direction is the direction that maximizes the path length through this high-emission region. For the purposes of the reconstruction algorithm, then, we can take the direction of the maximum soft X-ray emission to be the direction of the magnetopause tangent to a high degree of accuracy, at least for the geometries we have explored so far.

To further illustrate this principle, the lower panel in Figure 5 shows three line-of-sight cuts through this simulation originating from a satellite viewing at $(0, 0, 20) R_E$ in GSE coordinates. The blue line of sight cuts through the center of the subsolar magnetosheath. The red line of sight runs tangent to the magnetopause. The green line of sight intersects the subsolar magnetopause. The top panel of Figure 5 shows Q_{cut} along the line of sight for each of these three directions. At practically all distances from the spacecraft along the line of sight, the emission along the blue line of sight is lower than that along the red line of sight that runs tangent to the magnetopause, as expected because of the higher exospheric density along the tangent direction. Comparing the red tangent line of sight to the green line of sight that intersects the subsolar point, the emission along the subsolar line of sight usually exceeds that of the tangent line of sight within the magnetosheath. However, because the subsolar line of sight goes through the magnetosphere where there is no emission, the total integrated emission along the subsolar line of sight is lower than that along the tangent line of sight, as shown in the middle panel of Figure 5.

We also performed a similar analysis for a high solar wind flux case, $13.5 \times 10^8 \text{ cm}^2/\text{s}$, corresponding to a significantly more compressed magnetosphere. As in the case shown in Figure 5 for much lower solar wind flux, the maximum X-ray emission occurs along the line-of-sight tangent to the magnetopause with the emission along the subsolar line of sight resembling the green trace in which the depletion in emission along the portion of the line of sight that traverses the magnetospheric cavity results in a lower total integrated line-of-sight emission.

To apply the reconstruction algorithm discussed in the previous section, we must parametrize the spacecraft location by the tangent angle (i.e., the peak soft X-ray emission direction). Figure 6 illustrates this process. The top panel shows the variation of the X-ray peak angles (θ) observed by a single spacecraft in this circular orbit. In this circular orbit in the YZ plane, the satellite will observe multiple local X-ray peaks. The different colors in the top panel correspond to these different local peaks.

As the spacecraft moves through its orbit, it tracks the change in the local X-ray peak angle and thus traces the corresponding magnetopause location. However, the spacecraft eventually reaches a point where there no longer exists a geometric tangent line to the magnetopause position that has been traced up to that point. At this point, the local peak disappears.

The middle panel shows $|ds/d\theta|$ calculated along the spacecraft orbit. The colors in this panel correspond to the colors in the upper panel that represent the various local peaks in the soft X-ray intensity. The bottom panel shows the plasma density in the YZ plane (shading), the spacecraft orbit (black), and the magnetopause locations calculated from the reconstruction algorithm (red, blue, green, and cyan) and from the MHD model (yellow). The magnetopause locations determined from the reconstruction algorithm and the MHD

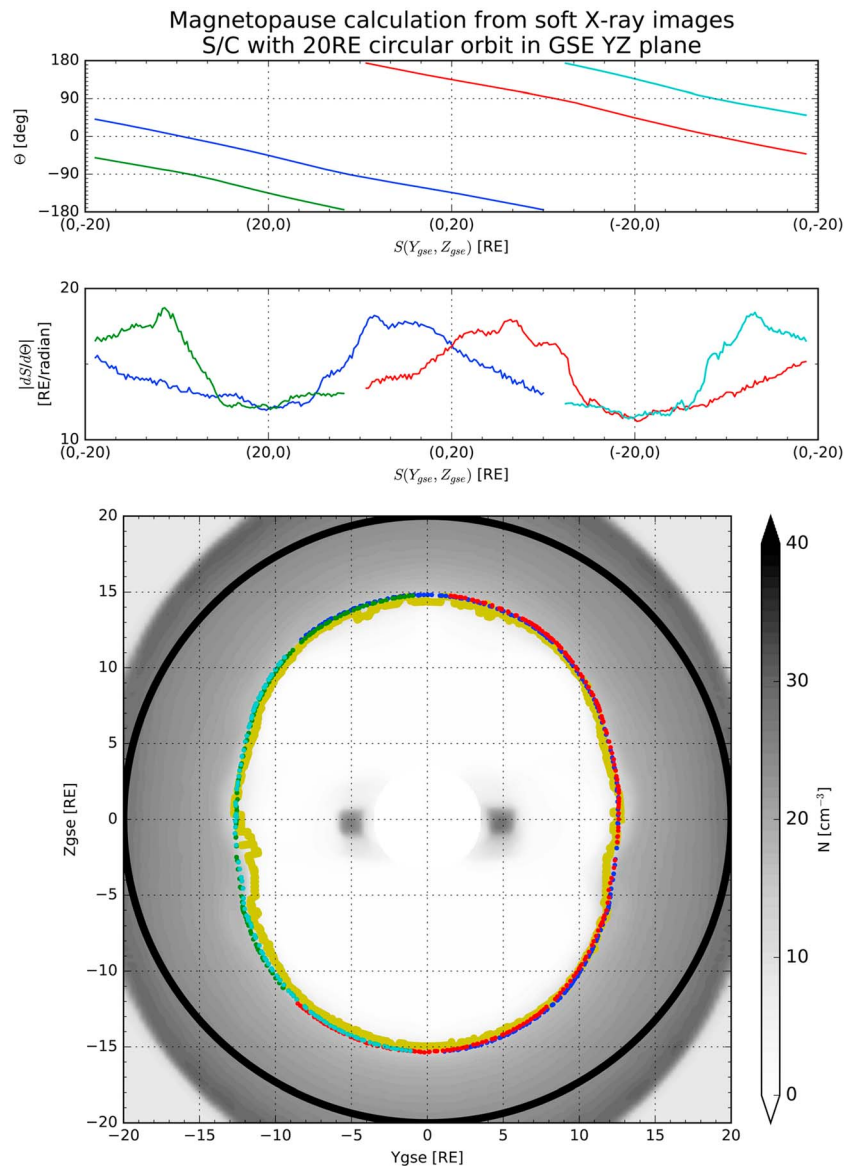


Figure 6. Application of the magnetopause reconstruction algorithm for a circular orbit in the GSE YZ plane (GSE $X = 0$). The colors relate the angles and their derivatives in the upper two panels to the observed tangent points on the magnetopause in the bottom panel.

simulation match very well. The red, blue, green, and cyan lines in Figure 6 represent the calculations from each X-ray peak angle.

Figure 7 shows the application of the reconstruction algorithm to a case in which the spacecraft executes a circular orbit in the GSE XZ plane of $20 R_E$ radius (black). Note that the algorithm is quite general and should work for any arbitrary orbit. The figure uses the same format as Figure 6: The top panel in Figure 7 shows the relationship between the spacecraft location, s , and the angle at which the maximum soft X-ray emission is observed, θ . The middle panel shows $|ds/d\theta|$, the derivative of the spacecraft position with respect to the observation angle of peak soft X-ray intensity, as determined from the data in the top panel. The bottom panel shows the magnetopause reconstruction resulting from the soft X-ray emission simulation (red, blue, and green) using equation (18) superimposed on the MHD plasma density results (shading). The black and yellow lines are the spacecraft orbit and the magnetopause locations calculated from the MHD model, respectively. As described earlier, the magnetopause location was

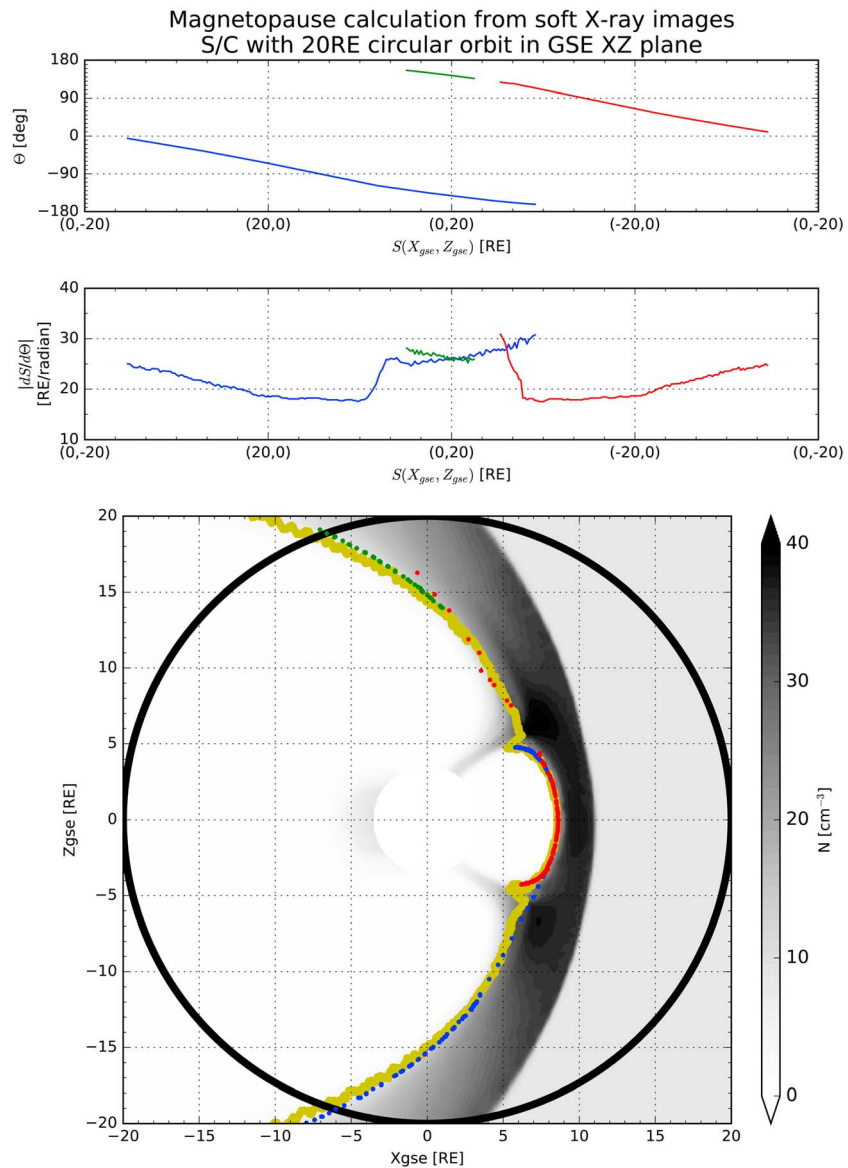


Figure 7. Application of the magnetopause reconstruction algorithm for a circular orbit in the GSE XZ plane (GSE $Y = 0$).

determined using the points along radial lines originating at the center of the Earth corresponding to the maximum density gradient.

The algorithm reconstructs the magnetopause location with reasonable accuracy. In fact, from some perspectives, the reconstruction algorithm manages to even reproduce the outer cusp shape as shown in the bottom panel of Figure 7.

These examples using an OpenGGCM MHD simulation show that, at least for these specific cases, the reconstruction algorithm, that is equation (18), accurately reproduces the magnetopause cross-sectional shapes from a variety of different spacecraft vantage points using the observed directions of the most intense soft X-ray emission.

4. Reconstruction Algorithm Uncertainties

One possible quantitative metric for evaluating the accuracy of the magnetopause reconstruction algorithm is the root-mean-square (rms) difference between the position of the magnetopause based on the MHD

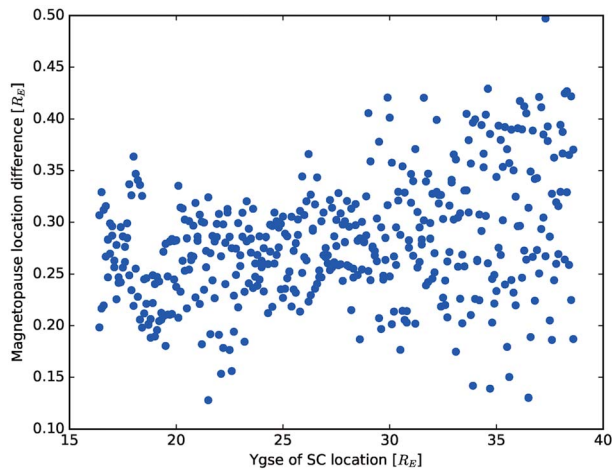


Figure 8. Error in the magnetopause location determined by the reconstruction algorithm as a function of spacecraft distance.

simulation and the position of the magnetopause calculated from the reconstruction algorithm. When this metric is applied to Figure 6, the GSE YZ plane cut at $X = 0$, the rms difference is $0.29 R_E$. When this metric is applied to Figure 7, the GSE XZ plane cut at $Y = 0$, the rms difference is $0.18 R_E$. The larger rms difference for the YZ plane in Figure 6 was due to the reconstruction algorithm not completely reproducing the earthward indentation in the southern magnetopause near $Z = -2$ to $-5 R_E$.

Figure 8 shows the difference between the magnetopause locations determined by the algorithm and those based on the MHD simulation for Figure 6, the GSE YZ plane cut at $X = 0$, as the satellite moves from about $40 R_E$ to about $15 R_E$. The reconstruction algorithm appears to be slightly more accurate closer to earth even though, as indicated earlier, the difference between the peak emission angle and the tangent angle to the magnetopause increases closer to Earth. This may occur because the angle changes more rapidly with satellite location closer to the Earth, which compensates for the difference between the peak emission angle and the tangent angle.

5. Three-Dimensional Reconstruction of the Magnetopause Surface

Given the results from the last section using a single spacecraft orbit to reconstruct the magnetopause surface in the plane of the orbit, it is clear that a precessing orbit will provide sufficient points on the magnetopause surface to generate a full three-dimensional reconstruction of the magnetopause.

Figure 9 shows the results of a surface fitted to the magnetopause points generated from 90 circular orbits over the poles with each orbit plane rotated two degrees with respect to the prior orbit plane (for a total of 42,654 points). The magnetopause surface was generated from these points via the *griddata* and *surf* functions in MATLAB using the *natural* interpolation method. The solar wind conditions in the simulation used to generate the data shown in Figure 9 are the same as in the simulations discussed in the last section, southward IMF of magnitude 5 nT.

The three-dimensional reconstruction shows some interesting structures, for example, depressions near the cusp locations. The three-dimensional reconstruction for the northward IMF case (not shown) exhibits both a less blunt shape and a subsolar point extending farther sunward than the southward IMF case, as expected (e.g., Fairfield, 1971).

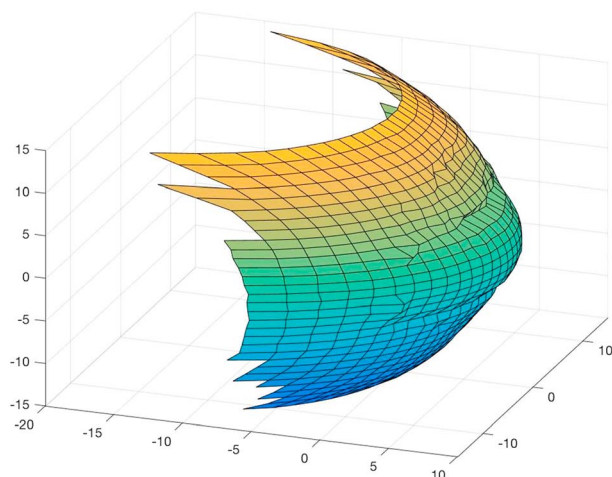


Figure 9. A full three-dimensional reconstruction of the magnetopause surface using a precessing spacecraft orbit.

Of course, solar wind conditions do not remain steady over the time required for a spacecraft in any orbit to precess through 180° . In practice, then, a reconstruction based on flight data similar to that shown in Figure 9 using simulations would most likely proceed by orbit binning the data according to the desired solar wind conditions and then reconstructing the magnetopause surfaces from the data subsets corresponding to these conditions, as magnetopause models tend to parametrize magnetopause shape using solar wind ram pressure and IMF B_z (e.g., Shue et al., 1997). How well magnetopause shape is characterized by solar wind parameters alone and what those parameters are will both be among the questions addressed by soft X-ray imaging. This also highlights the importance of accurately characterizing the solar wind conditions at the magnetopause (e.g., Collier et al., 1998; Richardson & Paularena, 1998) and suggests the optimal observational approach would outfit a high-apogee imaging spacecraft with its own in situ solar wind plasma and magnetometer suite.

6. Conclusions

This manuscript presented and validated through MHD simulation an algorithm for reconstructing the magnetopause surface, and more

generally any surface over which a spacecraft can observe tangent vectors using limb brightening. It should be pointed out that the algorithm does not require images, per se, only a measurement of the tangent vector direction so that, although the algorithm certainly could be used by imaging missions, the technique does not necessarily require an imaging mission. There are, however, limitations inherent in the reconstruction algorithm, which still need to be investigated fully (something that is beyond the scope of this paper), for example, the previously discussed assumption of time stationarity and the offset between the peak density gradient and the maximum soft X-ray emission.

Magnetopause reconstruction algorithms such as the one proposed here (see also Jorgensen et al. (2017)) are particularly timely because X-ray imaging mission concepts have been published (e.g., Branduardi-Raymont et al., 2012) and the upcoming joint European Space Agency-Chinese Academy of Sciences SMILE mission will perform soft X-ray imaging of the magnetosheath. We will soon be able to validate these magnetopause reconstruction algorithms with actual data.

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